NEURALPATH CHEATSHEET

Module 5: Sequence Models & Transformers

Go from BPE tokenization and embeddings through RNNs/LSTMs to self-attention, RoPE, GQA, BERT,

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Core Formulas, Mental Models & Engineering Rules

- BPE merges frequent adjacent character/byte pairs into compact subword vocabularies.
- Self-Attention: softmax((Q K^T) / sqrt(d_k) + M) V. Scale by sqrt(d_k) to prevent saturation.
- Modern Transformer: Pre-RMSNorm, RoPE positional rotations, GQA KV-cache sharing, SwiGLU.
- BERT uses bidirectional masked language modeling; GPT uses causal masking + KV caching.

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Estimated study time: 6 hours · Badge: Transformer Builder
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